Mandatory exercise 1

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Question 1

$$\begin{cases} -2w + 2x + 2y - 8z = 14 \\ 3w - 3x + y + 4z = -5 \\ 2w - 2x - 4y + 12z = -22 \end{cases}$$

Convert to matrix notation and reduce to reduced echelon form:

$$\left[\begin{array}{cccc|c} -2 & 2 & 2 & -8 & 14 \\ 3 & -3 & 1 & 4 & -5 \\ 2 & -2 & -4 & 12 & -22 \end{array}\right] \xrightarrow{R_1 \leftarrow -\frac{1}{2}R_1} \left[\begin{array}{cccc|c} 1 & -1 & -1 & 4 & -7 \\ 3 & -3 & 1 & 4 & -5 \\ 2 & -2 & -4 & 12 & -22 \end{array}\right]$$

$$\xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 3R_1} \left[\begin{array}{cccc|c} 1 & -1 & -1 & 4 & -7 \\ 0 & 0 & 4 & -8 & 16 \\ 0 & 0 & -2 & 4 & -8 \end{array}\right]$$

$$\xrightarrow[R_3 \leftarrow \frac{1}{2}R_3]{R_2 \leftarrow \frac{1}{4}R_2} \left[\begin{array}{cccc|c} 1 & -1 & -1 & 4 & -7 \\ 0 & 0 & 1 & -2 & 4 \\ 0 & 0 & -1 & 2 & -4 \end{array}\right]$$

$$\xrightarrow[R_3 \leftarrow R_3 + R_2]{R_1 \leftarrow R_1 + R_2} \left[\begin{array}{cccc|c} 1 & -1 & 0 & 2 & -3 \\ 0 & 0 & 1 & -2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The result is a system of 2 linearly independent equations with 4 unknowns, therefore the solution is a 2D plane in 4D space and the solution must have 2 linearly independent variables. First and third columns are pivot columns, second and fourth columns are free columns. Therefore the simplest way to find a solution is to set x and z to variables s and t and solving for w and y.

$$\begin{aligned} w &= s - 2t - 3 \\ x &= s \\ y &= 2t + 4 \\ z &= t \\ s, t &\in \mathbb{R} \end{aligned}$$

Question 2

Agriculture A mixture of 24 liters of chemical A, 32 liters of chemical B, and 52 liters of chemical C is required to kill a destructive crop insect. Commercial spray X contains 1, 2, and 2 parts, respectively, of these chemicals. Commercial spray Y contains only chemical C. Commercial spray Z contains chemicals A, B, and C in equal amounts. How much of each type of commercial spray is needed to get the desired mixture?

We need 24 liters, 32 liters and 52 liters of chemicals A, B and C respectively. These values are constant and therefore should go on the right of the vertical line in the augmented matrix. Each spray should occupy a column according to respective ratios (divided by the total amount of parts in the chemical spray) of the chemicals contained.

System of equations representation:

$$\begin{aligned}\frac{1}{5}X + 0Y + \frac{1}{3}Z &= 24\\ \frac{2}{5}X + 0Y + \frac{1}{3}Z &= 32\\ \frac{2}{5}X + \frac{1}{1}Y + \frac{1}{3}Z &= 52\end{aligned}$$

In order to prevent working with fractions, let's substitute:

$$\begin{aligned}x &= \frac{1}{5}X\\ z &= \frac{1}{3}Z\end{aligned}$$

Write substituted system of linear equations and convert to matrix notation:

$$\begin{aligned}x + 0Y + z &= 24\\ 2x + 0Y + z &= 32\\ 2x + Y + z &= 52\end{aligned}$$

$$\left[\begin{array}{ccc|c}1 & 0 & 1 & 24\\ 2 & 0 & 1 & 32\\ 2 & 1 & 1 & 52\end{array}\right]$$

The following subsections solve this matrix using different methods.

Using Gauss–Jordan elimination:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 24 \\ 2 & 0 & 1 & 32 \\ 2 & 1 & 1 & 52 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 24 \\ 0 & 0 & -1 & -16 \\ 0 & 1 & -1 & 4 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 24 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & -1 & -16 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 24 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & 16 \end{array}\right]$$

$$\xrightarrow[R_2 \leftarrow R_2 + R_3]{R_1 \leftarrow R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 16 \end{array}\right]$$

Substituting back to X and Z:

$$\left[\begin{array}{ccc|c} \frac{1}{5} & 0 & 0 & 8 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & \frac{1}{3} & 16 \end{array}\right]$$

Therefore to kill the destructive crop insect we need:

- 40 liters of commercial spray X
- 20 liters of commercial spray Y
- 48 liters of commercial spray Z

Using the transposition matrix:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

Generate the matrix of minors:

$$\begin{bmatrix} -1 & 0 & 2 \\ -1 & -1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

Generate the matrix of cofactors:

$$\begin{bmatrix} -1 & 0 & 2 \\ 1 & -1 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Find the adjoint matrix (transpose the matrix of cofactors):

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & -1 & 0 \end{bmatrix}$$

Divide the adjoint matrix with the determinant of the original matrix:

$$det(A) = \begin{vmatrix} 1 & 0 & 1 \\ 2 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -1 - 0 + 2 = 1$$

$$\frac{1}{1} \cdot \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & -1 & 0 \end{bmatrix}$$

Let:

$$\overrightarrow{x} = \begin{bmatrix} x \\ Y \\ z \end{bmatrix}, \overrightarrow{b} = \begin{bmatrix} 24 \\ 32 \\ 52 \end{bmatrix}$$

The following equation allows us to solve the problem using the transposition matrix:

$$A\overrightarrow{x} = \overrightarrow{b}$$

Multiply with A^{-1} from the left side to isolate $\overrightarrow{x}$

$$\begin{aligned} A^{-1}A\overrightarrow{x} &= A^{-1}\overrightarrow{b} \\ \overrightarrow{x} &= A^{-1}\overrightarrow{b} \end{aligned}$$

Find $\overrightarrow{x}$ by matrix multiplication:

$$\overrightarrow{x} = \begin{bmatrix} x \\ Y \\ z \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} 24 \\ 32 \\ 52 \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ 16 \end{bmatrix}$$

Substituting back to X and Z:

$$\begin{bmatrix} \frac{1}{5}X \\ Y \\ \frac{1}{3}Z \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ 16 \end{bmatrix}$$

Therefore to kill the destructive crop insect we need:

- 40 liters of commercial spray X
- 20 liters of commercial spray Y
- 48 liters of commercial spray Z

Using Cramer's rule:

First find the determinant of the original matrix.

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -1 - 0 + 2 = 1$$

Using Cramer's rule find the required amounts of x, Y and z using the explicit formula.

$$x = \frac{1}{1} \cdot \begin{vmatrix} 24 & 0 & 1 \\ 32 & 0 & 1 \\ 52 & 1 & 1 \end{vmatrix} = 8$$

$$Y = \frac{1}{1} \cdot \begin{vmatrix} 1 & 24 & 1 \\ 2 & 32 & 1 \\ 2 & 52 & 1 \end{vmatrix} = 20$$

$$z = \frac{1}{1} \cdot \begin{vmatrix} 1 & 0 & 24 \\ 2 & 0 & 32 \\ 2 & 1 & 52 \end{vmatrix} = 16$$

Substituting back to X and Z:

$$\frac{1}{5}X = 8$$
$$Y = 20$$
$$\frac{1}{3}Z = 16$$

Therefore to kill the destructive crop insect we need:

- 40 liters of commercial spray X
- 20 liters of commercial spray Y
- 48 liters of commercial spray Z

Question 3

(a) For which b does the equation $Ax = b$ have a solution? Draw your answer.

Let:

$$b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$
$$b_1, b_2 \in \mathbb{R}$$

Let's interpret and solve the problem as a system of linear equations.

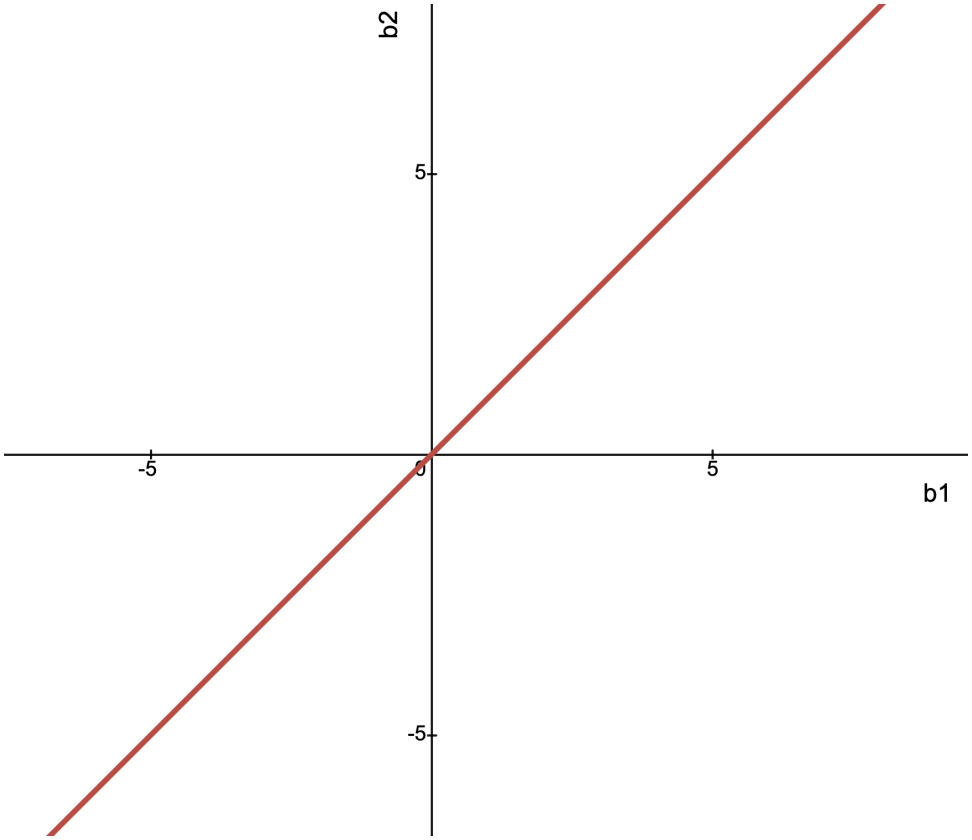
$$Ax = b$$
$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\frac{1}{2}x + \frac{1}{2}y = b_1$$
$$\frac{1}{2}x + \frac{1}{2}y = b_2$$
$$0 = b_2 - b_1$$
$$b_1 = b_2$$

Therefore the equation $Ax = b$ has a solution when b is of form

$$\begin{bmatrix} b_1 \\ b_1 \end{bmatrix}$$
$$b_1 \in \mathbb{R}$$

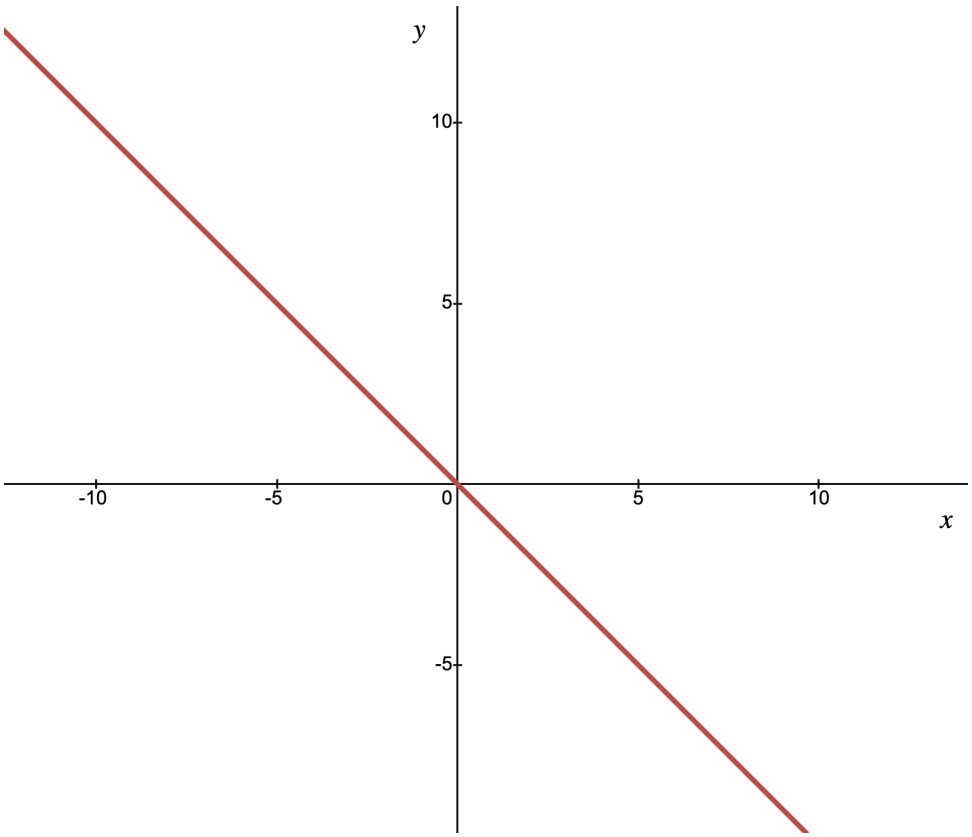
Graphically we find solutions in the linear combination of $\begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$ and $\begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$. Because these vectors are linearly dependent, their linear combination is a line.



(b) Which x satisfy the equation $Ax = 0$? Draw your answer

Just like in (a), let's interpret and solve the problem as a system of linear equations.

$$Ax = b$$
$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\frac{1}{2}x + \frac{1}{2}y = 0$$
$$x + y = 0$$
$$x = -y$$



(c) Compute the determinant of A.

$$det(A) = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4} = 0$$

(d) Is A invertible? Relate your answer to all the questions (a), (b) and (c) above.

Let's prove that A is not invertible using the invertible matrix theorem.
Invertible matrix theorem (simplified): Let A be an $n \times n$ matrix. The following statements are equivalent:

1. A is invertible.
2. $A\vec{x} = \vec{b}$ has a unique solution for all $\vec{b}$.
3. $A\vec{x} = \vec{0} \implies \vec{x} = \vec{0}$.
4. $\det(A) \neq 0$.

First clarify why statements 2.-4. are false for A.

2. From (a) we know that $A\vec{x} = \vec{b}$ does not have a solution for all $\vec{b}$. Instead it only has solutions in the special case, where $b_1 = b_2$.
3. $A\vec{x} = \vec{0} \implies \vec{x} = \vec{0}$. In (b) the premise is true and the conclusion is false for all solutions except $x = -y = 0$ therefore the implication must be false.

4. $\det(A) \neq 0$. Contradicts (c).

Because every equivalent statement to "A is invertible." is false, A is not invertible.

(e) (Not mandatory) If you can, describe in words what the function

$$f(x, y) = A \begin{bmatrix} x \\ y \end{bmatrix}$$

does, and relate it to your answer to question 3d above.

$f(x, y)$ is a linear transformation. Practically this means that it transforms space (in this case the 2D xy plane) such that [0, 0] maps to [0, 0] and grid lines remain parallel and evenly spaced.

Columns of A describe where the vectors $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ land after the transformation.

The determinant is the area of the parallelogram formed by the transformed $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ vectors and $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. $\hat{i}$, $\hat{j}$ and $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ land on the same line after the transformation, therefore no parallelogram can be formed and the determinant must be 0.

Linear transformations can be visualised for example on this Desmos site